

Ex. No.: 5

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System Calls Programming

Aim: To experiment system calls using fork(), execlp() and pid() functions.

Algorithm:

1. **Start**
 - Include the required header files (stdio.h and stdlib.h).
2. **Variable Declaration**
 - Declare an integer variable pid to hold the process ID.
3. **Create a Process**
 - Call the fork() function to create a new process. Store the return value in the pid variable:
 - If fork() returns:
 - -1: Forking failed (child process not created).
 - 0: Process is the child process.
 - Positive integer: Process is the parent process.
4. **Print Statement Executed Twice**
 - Print the statement:

scss

Copy code

THIS LINE EXECUTED TWICE

(This line is executed by both parent and child processes after fork()).

5. **Check for Process Creation Failure**

- If pid == -1:
 - Print:

Copy code

CHILD PROCESS NOT CREATED

- Exit the program using exit(0).

6. **Child Process Execution**

- If pid == 0 (child process):
 - Print:
 - Process ID of the child process using getpid().
 - Parent process ID of the child process using getppid().

7. **Parent Process Execution**

- If pid > 0 (parent process):
 - Print:
 - Process ID of the parent process using getpid().
 - Parent's parent process ID using getppid().

8. **Final Print Statement**

- Print the statement:

objectivec

Copy code
IT CAN BE EXECUTED TWICE

(This line is executed by both parent and child processes).

9. End

Program:

FORK SYSTEM CALL PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
int main ()
{
    int pid;
    pid = fork();
    printf ("\n THIS LINE EXECUTED TWICE");
    if (pid == -1) {
        printf ("\n CHILD PROCESS NOT CREATED\n");
        exit(0);
    }
    if (pid == 0) {
        printf ("\n I AM CHILD PROCESS AND MY ID IS %d\n", getpid());
        printf ("\n THE CHILD PARENT PROCESS ID IS: %d\n", getpid());
    }
    else {
        printf ("\n I AM PARENT PROCESS AND MY ID IS: %d\n", getpid());
        printf ("\n THE PARENTS PARENT PROCESS ID IS: %d\n", getpid());
    }
}
```

```
printf ("\n IT CAN BE EXECUTED TWICE");  
printf ("\n");
```

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Output:

THIS LINE EXECUTED TWICE
THIS LINE EXECUTED TWICE

I AM CHILD PROCESS AND MY ID IS: 12345
THE CHILD PARENT PROCESS ID IS: 12344.

I AM PARENT PROCESS AND MY ID IS: 12344
THE PARENTS PARENT PROCESS ID IS: 12343

IT CAN BE EXECUTED TWICE
IT CAN BE EXECUTED TWICE

Result:

Thus the system calls Program to experiment
system calls using fork(), execvp() and pid() ~~functions~~
has been executed successfully.

